Koteswara Rao Perumalla

India (Open to Relocation) | Applied AI Engineer | Backend Engineer | Full-Stack & ML Systems |
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EDUCATION

Bits Pilani, Dubai Campus — ***B.E, Computer Science*** ***Sep 2022 - Sep 2026***

EXPERIENCE

Trynocode ***Jul 2025 - Jan 2026***

Role - Software Developer

- Engineered full-stack features for Mabani (live PropTech app on Play Store), integrated 20+ **REST APIs** to feed real-time user data into backend AI automation workflows.
- Architected **Firebase** real-time sync listeners to handle event-driven triggers, cutting response latencies for active users.
- Implemented unit and API integration test suites, reducing system production bugs by ~25% ensuring high reliability.
- Contributed to development of an AI-powered notetaker app now listed on Amazon, engineering core features during early-stage product MVP builds.

Whizkey ***Jun 2024 - Aug 2024***

Role - Summer Intern *Remote*

- Built lead database of 250+ clients; wrote SEO-optimized content increasing organic traffic by ~20% over 2 months

PROJECTS

Clarity — Multi-Document RAG System + LangGraph Agent | FastAPI, LangGraph, LangChain, ChromaDB, Gemini, Next.js, TypeScript, Docker | Live: https://clarrity.vercel.app/

- Architected a production **RAG** pipeline supporting **multi-document ingestion** — PDF, DOCX, TXT, MD — with Gemini embeddings and **ChromaDB** vector store, enabling **semantic search** across multiple knowledge bases simultaneously
- Built a **LangGraph** ReAct agent with autonomous tool orchestration — document search, web search (DuckDuckGo), calculator, and datetime — agent decides which tool to use based on query context
- Implemented conversation memory via **LangGraph** MemorySaver with thread-based session management, enabling context-aware multi-turn conversations
- Implemented **SSE** streaming endpoint for real-time token-by-token agent responses, enabling ChatGPT-like typing experience
- Built LLM-based RAG evaluation pipeline scoring responses on faithfulness, relevance and completeness — enabling automated quality assessment
- Designed RESTful document management APIs — upload, list, delete by doc_id — with unique UUID tracking and chunk metadata per document
- Containerized full stack application with **Docker** and **Docker Compose** for reproducible local and production deployment

Expense Tracker (Full-Stack Web Application + AI Chatbot) | Tech Stack: FastAPI, Next.js, PostgreSQL, JWT, Authentication | Live: https://cashually.vercel.app/

- Integrated an **LLM**-powered **AI chatbot** enabling users to query and interact with their financial data through natural language, built on top of the existing **FastAPI** backend and **PostgreSQL** data layer.
- Architected a full-stack financial application using **FastAPI**, **Next.js**, and **PostgreSQL** to process secure, real-time user transactions.
- Designed 10+ RESTful API endpoints maintaining low-latency responses (<150ms), secured with **JWT** authentication and session controls.

AI Based Interview Assistant | Tech Stack: Python, Matplotlib, Hugging Face Transformer, Tkinter, Google Speech API, Language Tool.

- Engineered an automated response evaluation engine using fine-tuned **Hugging Face transformers** (**DistilBERT**) and vector-based **semantic search**, processing user inputs in <3 seconds with 80% sentiment alignment.
- Integrated **semantic similarity** algorithms (cosine distance) alongside keyword matching, improving candidate scoring accuracy by 15%.
- Architected automated performance reporting pipelines (visual heatmaps and telemetry), reducing manual evaluation overhead by an estimated 40%.

TECHNICAL SKILLS

Languages & Framework: Python, SQL, Java, C++, TypeScript, JavaScript, FastAPI, Flask, REST APIs, LangChain, LangGraph

AI/ML & Gen AI: PyTorch, TensorFlow, Scikit-Learn, Hugging Face (Transformers), LLMs, RAG Pipelines, ReAct Agent Pattern, Tool Orchestration, LoRA Fine-Tuning, Parameter-Efficient Fine-Tuning (PEFT), Prompt Engineering, Semantic Search, OpenRouter, Model Evaluation, YOLOv5, OpenCV, Sentiment Analysis.

Data, Databases & Infrastructure: PostgreSQL, ChromaDB (Vector DB), Supabase, Firebase, Vector Search, Pandas, NumPy, ETL Pipelines, Docker, Docker Compose, Git, CI/CD (basics)

Core Competencies: End-to-End AI System Development, Multi-Document RAG, Agentic AI Workflows, API Integration, Data-Driven Decision Making, Model Monitoring, A/B Testing (basics), Experiment Tracking